Duplicating DNA



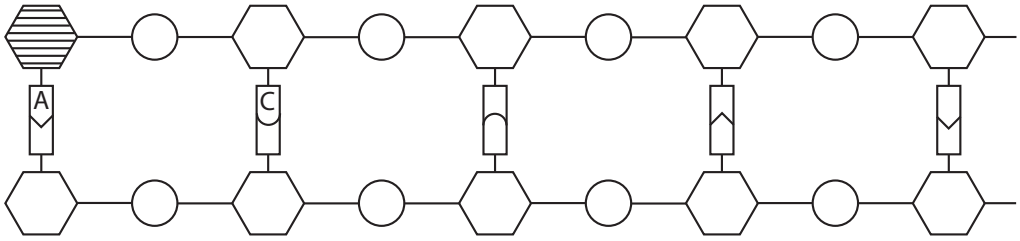
Chromosomes with their genes are made up of the chemical **DNA** or **deoxyribonucleic acid**. DNA is a long molecule that stores all genetic information. It looks like a double helix or a twisted ladder, as you can see in the diagram at the right.

DNA has only four repeated subunits, the **nucleotides**. In humans, a DNA molecule is about 2 cm long. In a 1 mm section of a DNA molecule there can be 8 million nucleotides. Each nucleotide is made up of a sugar unit, a phosphate unit and a base unit. The phosphate links the sugars to make the long strand (i.e. sugar – phosphate – sugar – phosphate – sugar). The bases of one strand lock onto the bases of the other strand to form the double strand.

Although the sugar and phosphate units are all the same, there are four kinds of bases: adenosine (A), cytosine (C), guanine (G) and thymine (T). The bases only join in certain pairs: **A – T** and **C – G**. In order to code for a particular gene such as tongue rolling, the DNA section of a particular chromosome is a certain length and the bases are in a particular order.

1 Identify the chemicals sugar, phosphate and the four bases of the DNA section by:
- shading **sugar** horizontally
- shading **phosphate** vertically
- labelling each base as **A, C, G** or **T**.

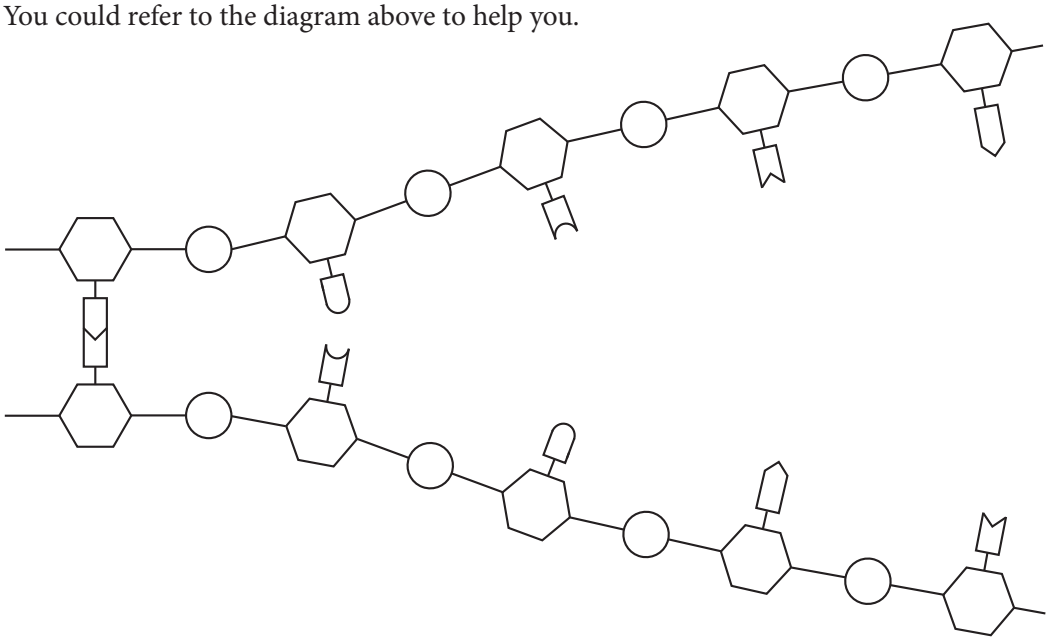
One sugar and two bases have been identified for you.



When a cell divides, it first has to make a copy of its DNA molecules. First, the double helix untwists. Then it splits open into two strands. The nucleotides which are floating around in the nucleus come into contact with the open strands and matching bases lock together. After a few minutes, two new double strands have been produced. It is important that the two new double strands are identical to the original.

2 The diagram below shows a DNA molecule splitting into two strands. For the parts that have already split:
- draw the nucleotides that will lock together with the open strands
- draw sugar and phosphate units to join the nucleotides you drew.

You could refer to the diagram above to help you.



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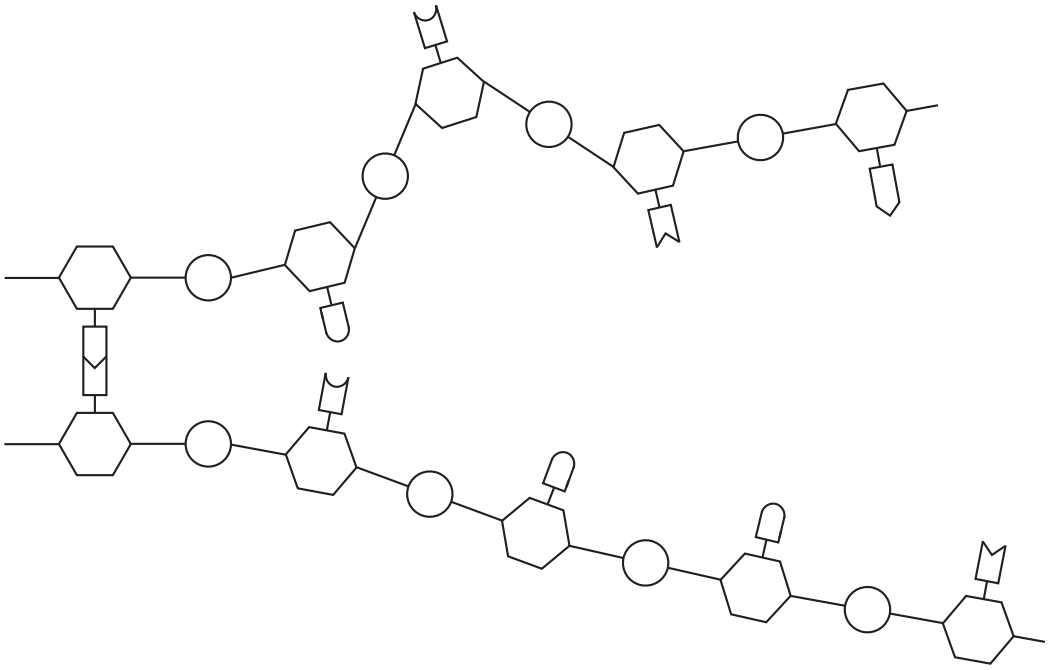
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3 Compare the new double strands with the original double strand. Are there any differences?

4 Compare the new double strands with each other. Are there any differences?

5 Explain how DNA duplication reduces the likelihood of errors.

Sometimes things go wrong during DNA duplication. In the diagram below, two mistakes have occurred.



6 As you did for question two, draw the double strands for the open sections of the diagram above. There will be two mistakes in the new double strands.

7 Describe the two mistakes in the new double strands.

Sometimes DNA can rectify small mistakes like those above. At other times, mistakes result in a new gene which creates a new characteristic in the organism. For example, it may no longer produce a certain enzyme which is needed during respiration, but produces a different enzyme instead. As a result, the cell will no longer be able to respire.

8 Explain what would happen to a cell if it could no longer respire.

Many DNA changes or mutations are harmless, such as a new gene that codes for a different eye colour. Some mutations are fatal and a few are beneficial. Mutations can be caused by radiation, certain chemicals and unusual heat.

9 Define *mutation*.

10 Explain why a pregnant woman should not have an X-ray.

11 If a pregnant woman takes medication without consulting her doctor, how might this affect the growing foetus?

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